

Gadded

AI-driven solar pre-development decision support for Egyptian factories

AI Empower Egypt 2026 - Renewable Energy Using AI | Technical Documentation

This document and the accompanying proof of concept describe a preliminary decision-support tool. Outputs are not legal, engineering, or financial advice; regulatory, engineering, grid, vendor, and financing details must be verified with the responsible authorities and qualified professionals.

1. Problem statement

Egypt is expanding renewable generation nationally, but individual factories and SMEs have no simple way to answer three connected questions before spending money on a feasibility study: **Can I legally and practically build a rooftop solar system here? Should I invest, and what size makes sense? What permits, approvals, and vendors do I need next?** Today this requires separately engaging a site surveyor, an energy engineer, a financial modeler, a regulatory consultant, and several EPC vendors before a factory owner even knows if the project is worth pursuing - an expensive and slow gate to a decision that renewable-energy adoption at scale requires many owners to make quickly and cheaply.

2. Motivation

Rooftop industrial solar is one of the fastest, lowest-conflict renewable deployments available in Egypt: it uses existing structures, adds no land-use conflict, and directly offsets grid-purchased electricity for energy-intensive sectors such as food processing and textiles. The barrier is informational, not technical - owners lack an accessible, trustworthy first pass at feasibility, sizing, economics, and process before committing to paid studies. Gadded is built to close that specific gap: a fast, evidence-grounded preliminary assessment that a factory owner can act on or safely discard.

3. Proposed approach and methodology

Gadded takes one assessment (location, sector, monthly electricity consumption, roof area, connection preference, financing preference) and runs it through eleven modules, each independently unit-tested, chained into one pipeline:

1. **Weather** - cached NASA POWER hourly irradiance and temperature for the site.
2. **Load profile** - an hourly factory consumption estimate, reconciled exactly to the submitted monthly kWh (see AI section below).
3. **PV generation** - hourly AC output from a physics simulation (`pvl`), not a data-driven model.
4. **Hourly matching** - self-consumed, imported, and exported energy for every hour of the year.
5. **Sizing optimization** - a constrained search over candidate capacities that maximizes project NPV, not simply the largest system the roof can hold.
6. **Finance** - deterministic cash and debt-financed scenarios (NPV, IRR, payback).
7. **Risk** - Monte Carlo simulation over PV yield, capex, opex, and tariff-escalation uncertainty, reusing the exact deterministic finance formulas per simulated draw.

8. **Site screening** - geospatial checks (industrial zone membership, protected-area intersection, nearest substation/road) against a local layer set.
9. **Regulatory screening** - retrieval over official excerpts, deterministic eligibility rules, and a cited plain-language explanation.
10. **Vendor discovery** - a web-search agent that returns only evidence-backed EPC leads.
11. **Feasibility & report** - a deterministic overall status and a combined, citation-bearing report.

The full pipeline runs end to end in one showcase notebook (`gadded.ipynb`) on a fixed golden case (a food-processing factory, day shift, 10th of Ramadan City) and is covered by over 90 automated tests (formula correctness, energy-balance reconciliation, evidence and citation completeness, reproducibility under a fixed random seed).

4. AI and technical components

Gadded deliberately separates six different kinds of computation and is explicit about which is which - a design requirement carried through the whole codebase, since conflating physics or optimization with "AI" would misrepresent the system.

Component	Category	What it does
Load-profile clustering	Machine learning	KMeans clustering (cluster count chosen by silhouette score) over a synthetic facility population, plus a RandomForest classifier mapping simple user inputs to a load-shape cluster. Evaluated on facilities held out at the facility level against a majority-lookup baseline; low-confidence predictions fall back automatically to a deterministic archetype.
PV generation	Physics simulation	<code>pvlib</code> plane-of-array transposition, cell temperature, and the PVWatts DC model - explicitly not described as ML anywhere in the codebase.
System sizing	Constrained optimization	Grid search over candidate capacities maximizing project NPV under roof-area and budget constraints.
Risk / uncertainty	Monte Carlo simulation	2,000 seeded draws over PV yield, capex, opex, and tariff escalation; one-at-a-time sensitivity ranking.
Regulatory retrieval	Retrieval	TF-IDF cosine similarity over short regulatory excerpts (no vector database needed at this corpus size).

Regulatory conclusion	Deterministic rules	Versioned eligibility rules evaluated with a safe, sandboxed condition language - never arbitrary code execution, and never decided by an LLM.
Regulatory explanation	LLM, grounded	Groq openai/gpt-oss-120b explains only the retrieved text and the rule conclusion already computed; it cannot invent a permit, threshold, authority, or duration.
Vendor discovery	LLM web-search agent	Groq groq/compound performs a real web search; a second grounded pass extracts strict, evidence-required vendor candidates. No quality ranking or unsupported claim is permitted; both stages fail soft.

5. Key assumptions, constraints, and design decisions

- **Scope:** industrial rooftop solar only, food-processing and textile sectors, one golden demo geography (10th of Ramadan City), self-consumption and net-metering connection models.
- **Every external or estimated value is versioned and classified** (OFFICIAL / MARKET_RANGE / LITERATURE_PROXY / SYNTHETIC / DEMO) in a single assumption registry.
- **Regulatory rules are grounded in real Egyptian law:** EgyptERA Circular No. 3/2023 and No. 6/2023 (self-consumption), Circular No. 4/2026 (net metering), Renewable Energy Law No. 203/2014, Protected Areas Law No. 102/1983, and Environmental Law No. 4/1994/EEAA guidance. The excerpt text is a paraphrased research summary of each circular's substantive requirements, not a verbatim legal quotation; wording and current amendments should be confirmed directly with EgyptERA/NREA/EEAA.
- **Site screening uses real geospatial data:** all 47 Egyptian protected areas from the World Database on Protected Areas (WDPA/WDOECM, UNEP-WCMC and IUCN), and the 10th of Ramadan City industrial-zone boundary, nearest substation, and nearest road from OpenStreetMap (via the Overpass API). One synthetic protected-area polygon is kept alongside the real WDPA layer solely to demonstrate the critical-finding code path in demos, and is clearly labeled as such.
- **Financial assumptions include real citations:** the official Egyptian medium-voltage industrial tariff bracket, and named bank solar-financing products (QNB Egypt Green Loan, Credit Agricole Egypt, CIB) for the financing rate, term, and fees. Capex is a real but consumer/commercial-market range (not a verified industrial EPC quote); export credit and the discount rate remain literature-proxy/demo values. The load-profile ML training population remains synthetic (no real Egyptian factory interval-metering data was available); the second (6th of October) industrial-zone layer also remains a synthetic placeholder.
- **The overall feasibility status is always decided by a fixed, ordered rule waterfall** - regulatory conclusions and site-screening severities in, one deterministic label out. The LLM is never in that decision path.
- **Financial results are always reported as a range** (Monte Carlo P10/P50/P90), never a single deterministic payback number.

- **A vendor or regulatory-LLM API failure is a warning, not a pipeline failure.** The technical, financial, and site results remain valid even if the live LLM calls are unavailable.

6. Results on the golden case

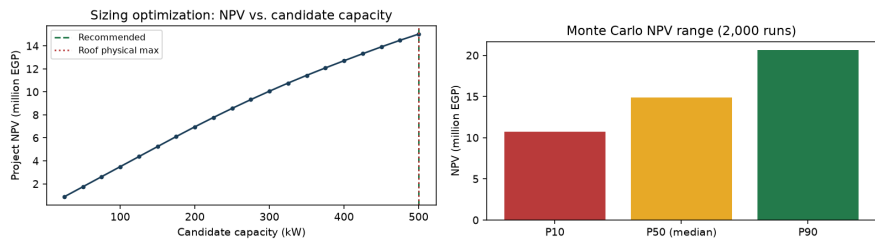
Golden case: a food-processing factory in the real 10th of Ramadan City industrial zone (Industrial Zone C6, coordinates verified against real OpenStreetMap boundaries), day shift, 12 monthly consumption values summing to **1,963,000 kWh/yr**, 3,000 m2 usable roof, owned, self-consumption connection, cash-vs-finance comparison, 6-year target payback.

Load-profile ML vs. baseline

Metric	Value
Cluster count (chosen by silhouette score)	6
Classifier test accuracy (facility-level held-out)	95.7%
Majority-lookup baseline accuracy	91.3%
Golden-case prediction confidence	high
Monthly reconciliation error	0.00%

The silhouette-selected cluster count (6) exactly matches the six known sector/shift archetypes used to generate the synthetic population - this validates the clustering/classification technique; it is not a claim of discovering hidden structure in real measured factories.

Recommended system and economics



Left: project NPV across candidate capacities - for this factory's consumption, real (much cheaper) capex data makes the roof's physical maximum the economic optimum. Right: Monte Carlo NPV range (2,000 runs).

Metric	Value
Recommended capacity	500 kW (= roof maximum; binding constraint: roof_area)
Annual PV generation	874,214 kWh
Self-consumption / self-sufficiency	90.0% / 40.1%
Capex	5,000,000 EGP
Year-1 net savings	2,006,755 EGP

NPV P10 / P50 / P90	10,735,896 / 14,844,303 / 20,642,210 EGP
Payback P10 / P50 / P90	2.4 / 2.7 / 3.1 years
Probability of meeting the 6-year target payback	100.0%
Top sensitivity driver	tariff escalation (68.2% of NPV sensitivity)
Overall feasibility status	feasible_with_conditions

With real, much lower capex data, the optimizer's economic optimum for this high-consumption factory is the full roof - the golden case does not exercise the "smaller than the roof" branch, which is instead proven by a dedicated unit test (a lower-demand scenario) and by the "oversized roof" preset in the demo UI. Every NPV percentile is comfortably positive and tariff escalation - not PV yield - dominates the remaining uncertainty; a real Egyptian industrial project also always requires an EEAA environmental-impact assessment (Circular 3/2023, Environmental Law 4/1994), which is why the status is "feasible with conditions" rather than "likely feasible" even on a clean, well-sited case.

7. Expected value and impact

For a factory owner, Gadded compresses a multi-week, multi-consultant feasibility process into one assessment that surfaces the same decision-relevant structure: a sized system with an economic rationale (not just "the biggest system that fits"), a financial range with its dominant uncertainty named, the concrete site and regulatory blockers to resolve next (grounded in the real EgyptERA/EEAA requirements that actually apply), and EPC leads with real evidence attached. Because every intermediate result carries its source, model version, and assumption classification, an owner or program officer can see exactly which numbers are provisional (DEMO) versus grounded, and a reviewer can trust the tool without trusting a black box. At a portfolio level, the same pipeline gives a government or program officer a consistent, repeatable way to screen many candidate projects for common deployment blockers before committing resources to detailed studies.

8. Limitations

- Regulatory excerpts are a paraphrased research summary of named EgyptERA circulars and Egyptian law, not verbatim legal text, and are not a substitute for confirming exact current wording with EgyptERA, NREA, and EEAA.
- GIS layers mix real sources (WDPA protected areas; OpenStreetMap industrial zone/road/substation for 10th of Ramadan City) with two remaining synthetic placeholders (the 6th of October zone, and one demo protected-area polygon kept only to exercise the critical-finding path) - both clearly labeled in the data itself.
- Capex is a real consumer/commercial-market range, not a verified industrial EPC quote; the discount rate and export-credit rate remain literature-proxy/demo values.
- The load-profile ML training population remains synthetic - no real Egyptian factory interval-metering data was available for this hackathon.
- Vendor and regulatory-LLM stages depend on a live third-party API and can degrade to "no result" under rate limiting or outage - by design, this never invalidates the technical/financial/site results.
- The proof of concept covers one project type (industrial rooftop), two sectors, and one geography; broader coverage is future work.

This is a preliminary decision-support assessment. Verify regulatory, engineering, grid, vendor, and financing information with the responsible authorities and qualified professionals before acting on any result.